European Polymer Journal 197 (2023) 112380

Contents lists available at ScienceDirect

European Polymer Journal

journal homepage: www.elsevier.com/locate/europolj

Enzyme assisted oxygen tolerance nitroxide-mediated miniemulsion polymerization

Xiaoguang Qiao [a,b,*], Xin Dong [b], Ge shi [b], Yanjie He [b], Xinchang Pang [b,*]

[a] College of Materials Engineering, Henan International Joint Laboratory of Rare Earth Composite Materials, Henan Engineering Technology Research Center for Fiber Preparation and Modification, Henan University of Engineering, Zhengzhou 451191, China

[b] Henan Joint International Research Laboratory of Living Polymerizations and Functional Nanomaterials, Henan Key Laboratory of Advanced Nylon Materials and Application, School of Materials Science and Engineering, Zhengzhou University, Zhengzhou 450001, China

A B S T R A C T

Nitroxide-mediated polymerization (NMP) is known to be a simple reversible deactivation radical polymerization (RDRP) method for preparing well defined polymers. In compared with other RDRP methods, NMP was highlighted by the clean and low toxic product which is favorable for some sensitive biological and electronic applications. Here we reported the first case of oxygen tolerance nitroxide-mediated miniemulsion polymerization, by introducing glucose oxidase (GOx) and ascorbic acid (AsAc) to the system. Without degassing treatment, the miniemulsion polymerization can be carried on using alkoxyamine 3-(((2-cyanopropan-2-yl)oxy) (cyclohexyl)amino)-2,2-dimethyl-3-phenylpropanenitrile (Dispolreg 007) as the initiator. The oxygen tolerance NMP miniemulsion exhibited similar living and control behavior to the degassing system, and the chain extension could also be performed without degassing. The characteristics of this enzyme assisted oxygen tolerant heterogeneous NMP may provide exciting new opportunities for increased versatility of NMP polymerization in preparation of polymeric nano-objects.

1. Introduction

As a typical heterogeneous system, miniemulsion are nano-sized oil droplets (50–500 nm) in water [1,2]. Compared with other techniques, miniemulsion polymerization has certain advantages such as high heat transfer efficiency, low viscosity and highly relevant for environmentally friendly characters, and is therefore widely used for the production of various polymer microspheres [3,4]. The reversible deactivation radical polymerization (RDRP) techniques including atom transfer radical polymerization (ATRP) [5–10], reversible addition fragmentation chain transfer (RAFT) [11–16] and nitroxide mediated polymerization (NMP) [17–20] had also been applied to produce polymers through miniemulsion polymerization. Owing to the significant advantages of RDRP, the achieved polymer exhibited confined molar mass, controlled molar mass distribution, architecture and functionality. Furthermore, electrochemically and photo mediated RDRP techniques were also employed for miniemulsion polymerization, allowing polymerization to be carried on under more milder conditions and facile "ON/OFF" control [21–27].

Recently, the development of the RDRP focus on oxygen tolerance techniques which are particularly important for some special polymerization systems such as in vivo and high-throughput systems that should be performed under aerobic conditions. Great progress has been made for oxygen tolerance RAFT and ATRP [28–40]. The direct oxygen tolerance performance was consuming the oxygen in the system, and various of oxygen consumers were therefore developed for oxygen resistant RDRP. For example, glucose oxidase (GOx) enzymes have been utilized as oxygen consumer for both thermal RAFT polymerization [33,36,39] and photo-ATRP in aqueous media [30,31]. Moreover, reducing agents [35] and the method so called "through oxygen polymerization" [28,30,32] were also utilized as alternative strategies for oxygen resistant RDRP. Recently, our group reported the first case of oxygen tolerance NMP with alkoxyamine 3-(((2-cyanopropan-2-yl)oxy) (cyclohexyl)amino)-2,2-dimethyl-3-phenylpropanenitrile (Dispolreg 007) as the initiator [41]. The polymerization can be performed without any common deoxidation method, relying solely on the special structure of Dispolreg 007. So far, some works have reported the oxygen tolerance emulsion or miniemulsion polymerization through ATRP and RAFT techniques, but oxygen tolerance NMP miniemulsion polymerization has rarely been reported to the best of our knowledge.

Here we report the first case of oxygen tolerance NMP miniemulsion polymerization, with Dispolreg 007 as the initiator, n-butyl methacrylate (BMA) or benzyl methacrylate (BzMA) as the monomers. GOx and ascorbic acid (AsAc) were used in conjunction to form oxygen

* Corresponding authors at: College of Materials Engineering, Henan International Joint Laboratory of Rare Earth Composite Materials, Henan Engineering Technology Research Center for Fiber Preparation and Modification, Henan University of Engineering, Zhengzhou 451191, China (X. Pang).

E-mail addresses: qiaoxiaoguang@haue.edu.cn (X. Qiao), Pangxinchang1980@163.com (X. Pang).

https://doi.org/10.1016/j.eurpolymj.2023.112380
Received 7 June 2023; Received in revised form 31 July 2023; Accepted 20 August 2023
Available online 24 August 2023
0014-3057/© 2023 Elsevier Ltd. All rights reserved.

Table 1
Table of conditions and main characteristic for NMP miniemulsion polymerization.[a]

No	Oxygen	Monomer	Surfactant (wt%)	AsAc (mg/ml)	ratio[b]	Conv,[c] (%)	$M_{n,th}$[d] ($g.mol^{-1}$)	$M_{n,GPC}$[e] ($g.mol^{-1}$)	M_w/M_n
1	D[f]	BMA	4	/	200:1	68.6	19,500	16,000	1.54
2	WD[g]	BMA	4	/	200:1	/	/	/	/
3	WD	BMA	4	4	200:1	85.6	21,200	31,500	1.68
4	WD	BMA	4	2	200:1	77.4	16,600	30,100	1.69
5	WD	BMA	4	6	200:1	90.1	23,700	35,400	1.83
6	WD	BMA	2	4	200:1	79.9	23,200	33,200	1.65
7	WD	BMA	6	4	200:1	89.5	17,900	30,300	1.72
8	WD	BMA	4	4	100:1	90.2	11,600	17,700	1.76
9	WD	BMA	4	4	300:1	75.3	30,300	48,100	1.70
10	WD	BzMA	4	4	200:1	66	25,100	34,300	1.69

[a] Reaction condition: 20% solid content, 90 ℃. Reaction time: 2.5 h.
[b] The ratio of monomer to Dispolreg 007.
[c] Monomer conversions were determined by mass method.
[d] Theoretical molecular weight was calculated by the following equation: $M_{n,theo} = [M]_0/[I]_0 \times M_W^{Monomer} \times \alpha + M_W^{Initiator}$.
[e] Molecular weight and polydispersity were determined by gel permeation chromatography (GPC) using THF as eluent calibrated to polystyrene standards.
[f] "D" means "degassing".
[g] "WD" means "without degassing".

consuming environment for minimeulsion polymerization. The kinetic behavior of this enzyme assisted oxygen tolerance NMP minimeulsion polymerization was fully studied. We also discussed the effect of AsAc and surfactant contents on the polymerization, as well as the chain extension without degassing.

2. Experimental section

2.1. Materials

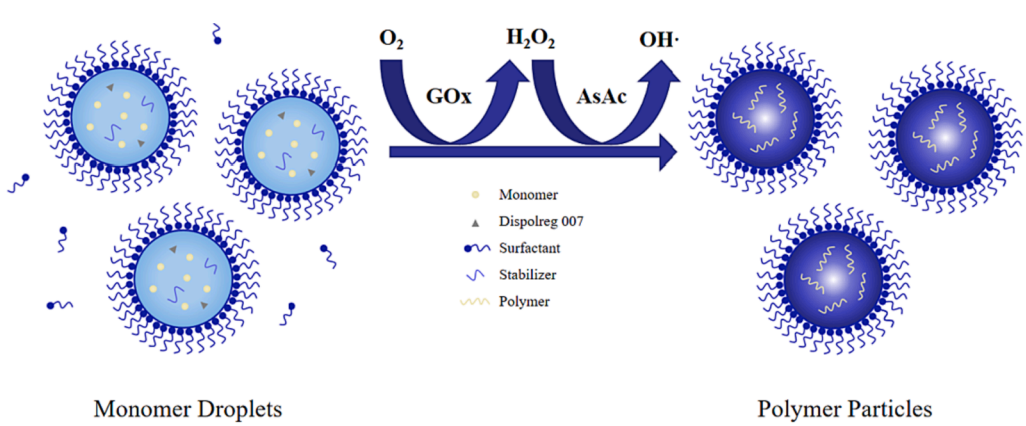
Unless otherwise specified, all chemical reagents were purchased and used as received. n-butyl methacrylate (BMA) and benzyl methacrylate (BzMA) were passed over basic aluminum oxide to remove inhibitors before using. Benzyl bromide, cyclohexylamine, 2,2′-azobis(2-methylpropionitrile) (AIBN), isopropanol, n-heptane, ethyl acetate, and the 4-amino-2,2,6,6-tetramethylpiperidine 1-oxyl free radical (NH_2-TEMPO), Disodium 4-dodecyl-2,4′-oxydibenzenesulfonate (MADS), hexadecane (HD), D-(+) glucose, glucose oxidase (GOx) enzymes and ascorbic acid (AsAc) were purchased from Aladdin Chemistry. s-Trioxane was purchased from Macklin Chemistry. Potassium peroxymonosulfate (Oxone®) was purchased from Sigma Aldrich. Sodium bicarbonate ($NaHCO_3$), acetone, N,N-dimethylformamide (DMF) and toluene were purchased from Tianjin Kaitong Chemical Reagent Company. Deionized water was utilized for all the polymerization.

2.2. Measurement

^{1}H NMR spectra were recorded on Bruker 400 MHz spectrometer with $CDCl_3$ as the solvent. Molar mass and molar mass distributions were characterized using a gel permeation chromatography (GPC) system equipped with a Waters 1515 HPLC pump and a refractive index detector (RID-0.65 A, 220 V). THF was used as the mobile phase with a flow rate of 1.0 mL min^{-1} under 35 ℃, and PMMA standard was selected to be the calibration standard. The mean size and size distribution of the upconversion nanoparticles were measured by dynamic light scattering (DLS) using Nanotrac wave II at 298° K. The size and morphology of the nanoparticles were observed using transmission electron microscopy (TEM, CM 200) at an accelerating voltage of 120 kV, after staining with phosphotungstic acid.

2.3. Enzyme assistant NMP miniemulsion polymerization

Miniemulsion polymerizations were carried out in a 10 mL Schlenk flask. In a typical reaction (Table S1), HD (acting as a costabilizer) (0.2 g) and Dispolreg 007 (0.0204 g, 0.034 mmol) were dissolved in BMA (1 g, 6.875 mmol). The organic phase was added to a stirred solution of MADS (0.03 g) and glucose (0.27 g) in water (1.5 g). The coarse emulsion was stirred for 15 min and then sonicated using a Scientz-750F sonicator operating at 70% output control and 0.5 duty cycle for 5 min in an ice bath. Subsequently, 0.04 g AsAc and 2 μM GOx were added to the miniemulsion. After stirring 10 min, the temperature was increased to 90℃ to start the polymerization. Samples were taken



Scheme 1. Enzyme assisted oxygen tolerance NMP Miniemulsion polymerization.

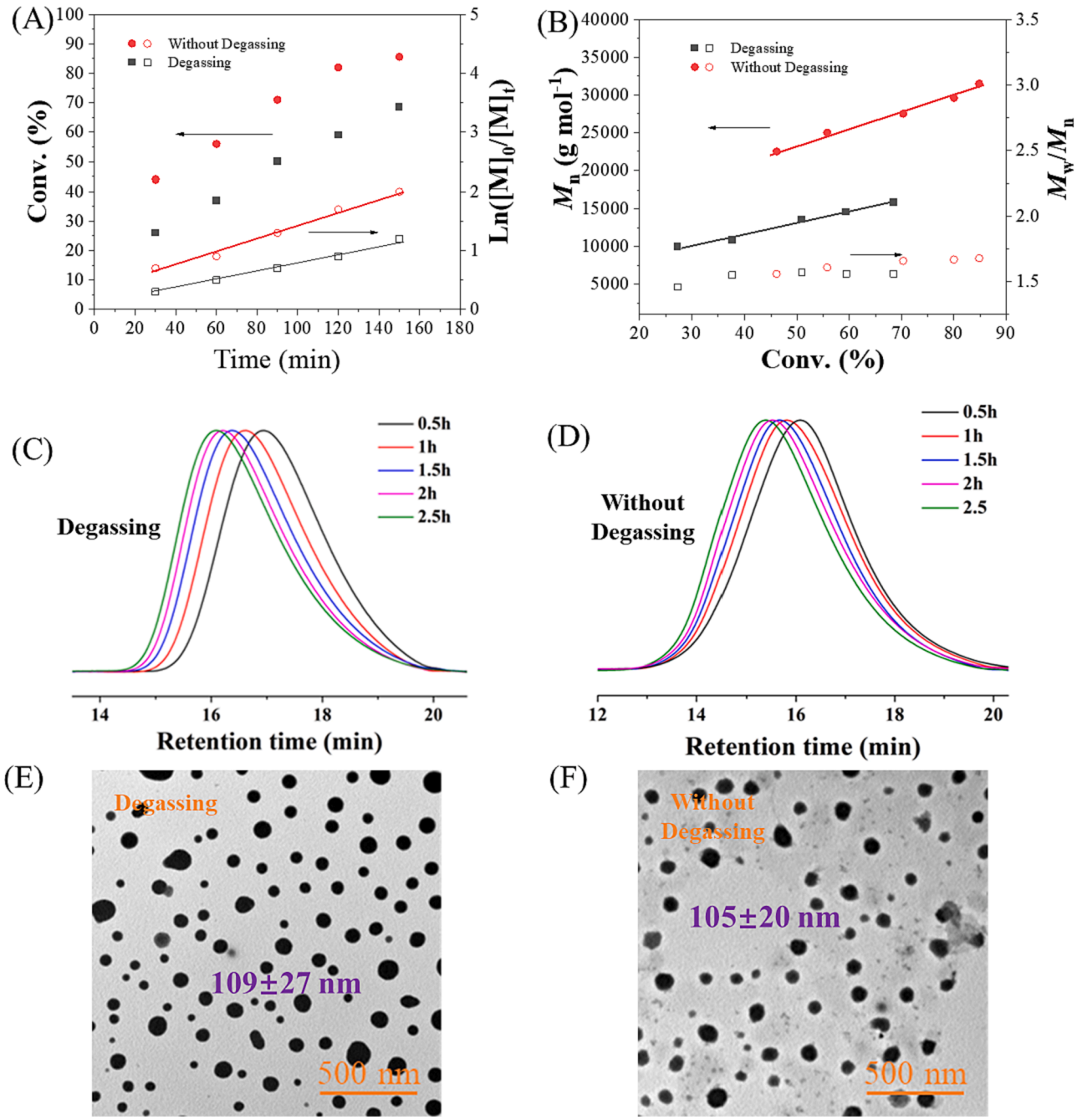


Fig. 1. Dispolreg 007 initiated the NMP miniemulsion polymerization of BMA. (A) Kinetic plot of monomer conversion and the Ln($[M]_0/[M]_t$) versus time for polymerizations with and without degassing. (B) Evolution of M_n and M_w/M_n with monomer conversion for polymerizations with and without degassing. (C) and (D) GPC traces of polymerization with and without degassing treatment. (E) and (F) TEM images of polymer latex particles for the polymerization with and without degassing treatment (stained by phosphotungstic acid.).

periodically for the measurement of the molecular weight and conversion.

3. Results and discussion

3.1. Oxygen tolerance NMP miniemulsion polymerization

3-(((2-cyanopropan-2-yl)oxy)-(cyclohexyl)amino)-2,2-dimethyl-3-phenylpropanenitrile (Dispolreg 007) was synthesized according to procedure shown in Figure S1, referring to the works of Asua and our previous works [41–44]. We first attempted oxygen tolerance NMP miniemulsion polymerization with only Dispolreg 007 (entry 2 in Table 1), but it seems that the solution oxygen tolerance mechanism was not suitable for miniemulsion system. In a miniemulsion system, the monomer is dispersed in the water in the form of a large number of small droplets which greatly increased the contact area of monomer with oxygen. According to our previous work, Dispolreg 007 cannot consume large amount of oxygen which similar to NMP solution system that cannot be carried on under "open to air" condition [41]. Therefore, we developed an enzyme assisted oxygen resistant NMP miniemulsion polymerization system as described in Scheme 1. GOx was used to oxidize glucose, and formed hydrogen peroxide (H_2O_2) which then reacted with AsAc through redox reaction to produce hydroxyl radicals. Oxygen free conditions for the polymerization were achieved through

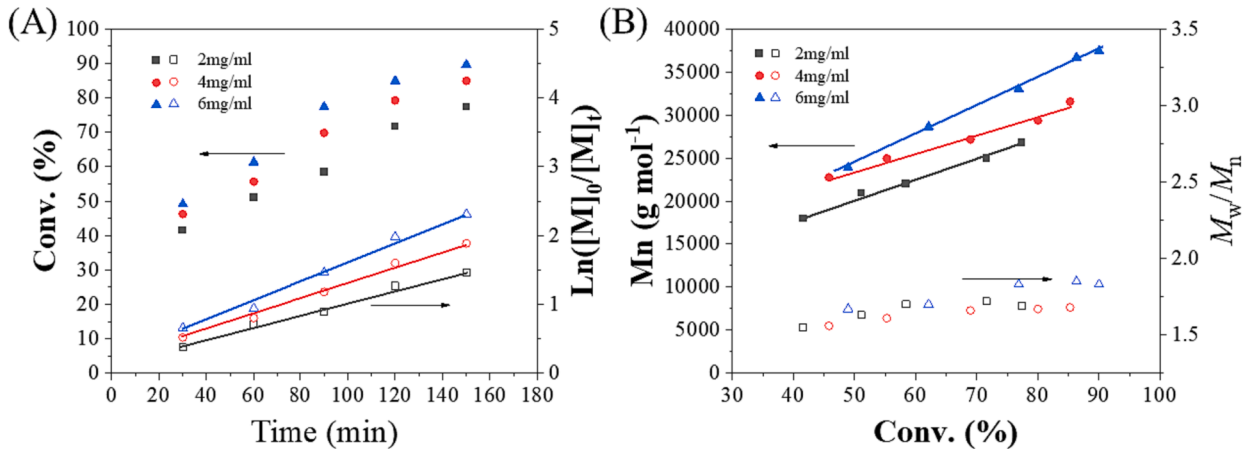


Fig. 2. Enzyme Assisted oxygen tolerance NMP miniemulsion polymerization with different amounts of AsAc. (A) Kinetic plot of monomer conversion and the Ln ([M]0/[M]t) versus time. (B) Evolution of M_n and M_w/M_n with monomer conversion for polymerizations.

the consumption of oxygen by GOx and hydroxyl radicals [23].

Using BMA as the monomer and Dispolreg 007 as the initiator, we conducted the NMP miniemulsion polymerization deoxygenated by nitrogen bubble method (D) and the enzyme assisted polymerization without degassing (WD), respectively. Both of the miniemulsion system with and without degassing showed the linear semi-logarithm curve fitting for $\ln([M]_0/[M]_t)$ *vs* time in Fig. 1A, indicating the stable propagating radical concentration in the systems. Moreover, the enzyme assisted system showed a faster polymerization rate than degassing system. The monomer conversion reached 85.6% within 150 min for the enzyme assisted system, while it was 68.6% for the degassing system within the same reaction time. The rate increase was ascribed to the reaction of AsAc with free nitroxides which shifts the equilibrium to the active state, and induced higher propagation radical concentration in the system [45]. Besides the polymerization rate, higher propagation radical concentration also showed some effect on the molar mass distribution. As shown in Table 1 (Entry 3), the molar mass distribution for the final polymer of enzyme assisted system was 1.68, which is a little greater than degassing system (Entry 1, $M_w/M_n = 1.54$). Fig. 1B showed that the molecular weight linear increased with the monomer conversion for both polymerizations with and without degassing, which confirmed the living character of the NMP miniemulsion polymerization. The good control of degassing and enzyme assisted polymerization were also proved by the mono-shift of the GPC traces (Fig. 1C and D). The Z-average particle sizes of the two particles measured by DLS were 167 nm (D) and 165 nm (WD), and the polydispersion indexes (PDI) of the particles were 0.13 (D) and 0.15 (WD), respectively. It seems that the enzyme did not affect the emulsion stability of NMP miniemulsion polymerization. The results were also proved by TEM images (Fig. 1E and F), in which the diameters of particles were 109 ± 27 and 105 ± 20 nm, respectively.

3.2. *Effect of AsAc and surfactant on the NMP miniemulsion polymerization*

Since the addition of AsAc resulted in the formation of hydroxyl radicals and reaction with free nitroxide to accelerate the polymerization, we therefore added different amounts of AsAc to the miniemulsion system (Table 1, Entry 3, 4 and 5). As shown in Fig. 2A, the polymerization rate increased with the increase of AsAc content, because the propagation radical concentration was increased as described above. Although the molecular weight linear increased with the monomer conversion for both polymerizations with different AsAc contents, the molar mass distributions became greater for higher AsAc contents (Fig. 2B).

As listed in Table 1 (entries 3, 6 and 7), the amount of surfactant (SDS) was varied from 2.0 to 6.0 wt% based on monomer. Higher polymerization rate was achieved for larger SDS concentration, which resulted in smaller monomer droplets and therefore larger number concentration (Table S2 and Fig. 3A). In the case of RDRP miniemulsion polymerization, the so-called segregation effect (compartimentalization) may lead to a reduction in the rate of bimolecular termination. Compared with the polymerization rate, the amount of surfactant has less effect on the control of molar mass and molar mass distributions which were close to 1.65 (Fig. 3B).

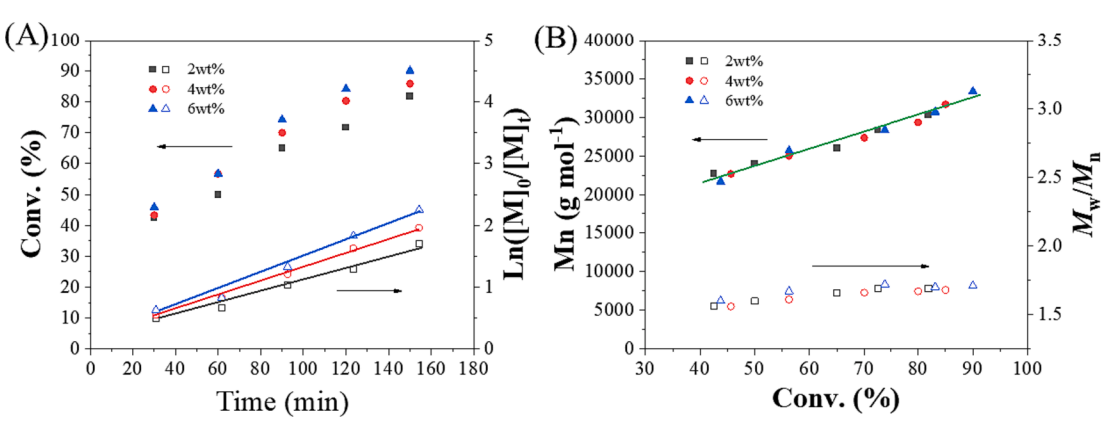


Fig. 3. Enzyme Assisted oxygen tolerance NMP miniemulsion polymerization with different amounts of surfactant. (A) Kinetic plot of monomer conversion and the $\ln([M]_0/[M]_t)$ versus time. (B) Evolution of M_n and M_w/M_n with monomer conversion for polymerizations.

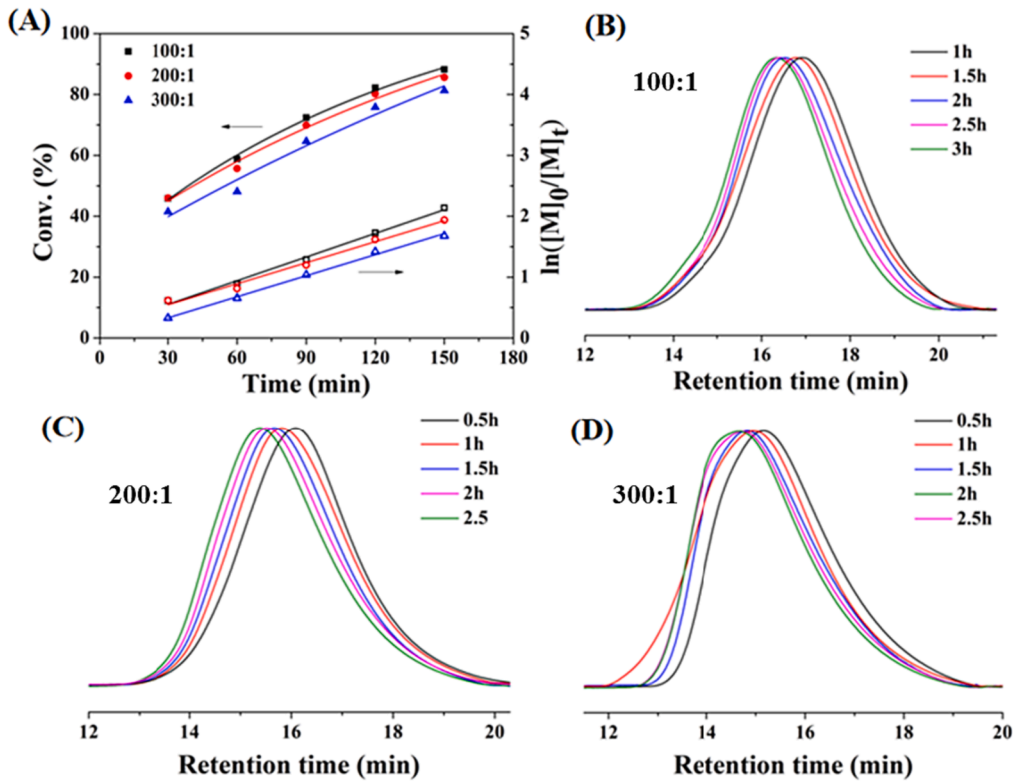


Fig. 4. Dispolreg 007 initiated the NMP miniemulsion polymerization of BMA for different targeting degrees of polymerization (DPs). (A) Kinetic plot of monomer conversion and the conversion of polymerization with time. (B, C, D) GPC traces of NMP polymerization for BMA in miniemulsion with different ratio of [monomer]:[Dispolreg 007].

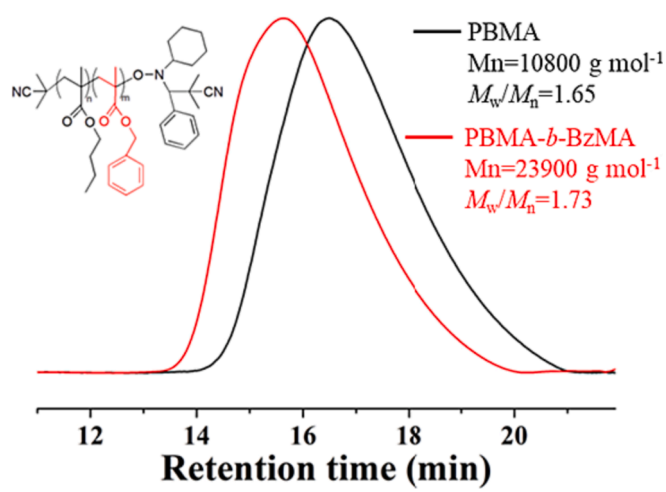


Fig. 5. Chain extension of macroalkoxyamine PBMA with monomer BzMA through enzyme assisted NMP minimemulsion polymerization process.

3.3. Versatility study of the enzyme assisted oxygen tolerance NMP minimeulsion polymerization

Different target DPs (Table 1, Entry 3, 8 and 9) were set to discuss the versatility of this enzyme assisted oxygen tolerance NMP minimeulsion polymerization system. As shown in Fig. 4A, the linear semi-logarithm curves fitting for $\ln([M]_0/[M]_t)$ *vs* time were observed for polymerization with different target DPs. The polymerization rated decreased with the increased of target DPs as expected, which related to lower alkoxyamine concentration due to the fixed monomer concentration. The good control and livingness of the polymerizations with different target DPs were also proved by the mono-shift of the GPC traces (Fig. 1B to D). The enzyme assisted NMP miniemulsion polymerization was also performed with BzMA as the monomer. As shown in Table 1 (Entry 10), polymerization with BzMA reached 66% monomer conversion within 2.5 h and polymer with molar mass of 25100 kg mol^{-1} ($M_w/M_n = 1.69$) was achieved.

3.4. Chain extension in oxygen tolerance NMP minemulsion polymerization

Chain extension was performed with a PBMA macroalkoxyamine ($M_n = 10800$ g mol^{-1}, $M_w/M_n = 1.65$) achieved from enzyme assisted oxygen tolerance NMP minimeulsion polymerization, and BzMA as the second monomer. The macroalkoxyamine was first dissolved in BzMA and then applied to form miniemulsion system. Through the same oxygen tolerance NMP miniemulsion polymerization, PBMA-*b*-PBzMA with a molecular weight of 23900 g mol^{-1} ($M_w/M_n = 1.73$) was got. As shown in Fig. 5, the mono-shifted GPC traces illustrated the high livingness of macroalkoxyamine.

4. Conclusion

In conclusion, with the assistant of GOx and AsAc, NMP miniemulsion polymerization could be performed without degassing treatment. With joint participation of GOx and AsAc, hydrogen peroxide was formed to consume the oxygen. The Enzyme Assisted oxygen tolerance NMP miniemulsion polymerization exhibited good living and control behavior, which was similar to the degassing system. Moreover, both higher AsAc contents and larger SDS concentrations would induce higher polymerization rate. In addition, chain extension could also be performed in this enzyme assisted oxygen tolerance NMP miniemulsion polymerization system. The oxygen tolerance miniemulsion polymerization system may provide exciting new opportunities for increased

versatility of NMP polymerization in preparation of polymeric nano-objects.

CRediT authorship contribution statement

Xiaoguang Qiao: Conceptualization, Formal analysis, Investigation, Funding acquisition, Writing - original draft, Writing - review & editing. **Xin Dong:** Investigation, Data curation. **Xinchang Pang:** Supervision, Resources, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

The financial support from the National Natural Science Foundation of China (Grant No.51973201, U1804128, to Xinchang Pang; Grant No.52173209, to Yanjie He), the National Science Foundation for Young Scientists of China (Grant No. 22105179, to Ge Shi) and Scientific & technological research projects in Henan Province (222102520009, to Xiaoguang Qiao) are greatly appreciated.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.eurpolymj.2023.112380.

References

[1] J.M. Asua, Miniemulsion polymerization, Prog. Polym. Sci. 27 (7) (2002) 1283–1346.
[2] J.M. Asua, Challenges for industrialization of miniemulsion polymerization, Prog. Polym. Sci. 39 (10) (2014) 1797–1826.
[3] A. Gupta, H.B. Eral, T.A. Hatton, P.S. Doyle, Nanoemulsions: formation, properties and applications, Soft matter. 12 (11) (2016) 2826–2841.
[4] Y. Singh, J.G. Meher, K. Raval, F.A. Khan, M. Chaurasia, N.K. Jain, M.K. Chourasia, Nanoemulsion: Concepts, development and applications in drug delivery, J. Control Release 252 (2017) 28–49.
[5] C. Lv, C. He, X. Pan, Oxygen-Initiated and Regulated Controlled Radical Polymerization under Ambient Conditions, Angew. Chem., Int. Ed. 57 (30) (2018) 9430–9433.
[6] A.M. Elsen, J. Burdyńska, S. Park, K. Matyjaszewski, Active ligand for low PPM Miniemulsion Atom Transfer Radical polymerization, Macromolecules 45 (18) (2012) 7356–7360.
[7] K.O. Jung, D.J. Siegwart, H.I. Lee, G. Sherwood, L. Peteanu, J.O. Hollinger, K. Kataoka, K. Matyjaszewski, Biodegradable Nanogels Prepared by Atom Transfer Radical Polymerization as Potential Drug Delivery Carriers: Synthesis, Biodegradation, in Vitro Release, and Bioconjugation, J. Am. Chem. Soc. 129 (18) (2007) 5939–5945.
[8] K. Matyjaszewski, J. Qiu, N.V. Tsarevsky, B. Charleux, Atom transfer radical polymerization of n-butyl methacrylate in an aqueous dispersed system: A miniemulsion approach, J. Polym. Sci., Part A: Polym. Chem. 38 (S1) (2000) 4724–4734.
[9] K. Min, H. Gao, K. Matyjaszewski, Preparation of Homopolymers and Block Copolymers in Miniemulsion by ATRP Using Activators Generated by Electron Transfer (AGET), J. Am. Chem. Soc. 127 (11) (2005) 3825–3830.
[10] Y. Kitayama, M. Yorizane, Y. Kagawa, H. Minami, P.B. Zetterlund, M. Okubo, Preparation of onion-like multilayered particles comprising mainly poly(iso-butyl methacrylate)-block-polystyrene by two-step AGET ATRP, Polymer 50 (14) (2009) 3182–3187.
[11] T. Boursier, I. Chaduc, J. Rieger, F. D'Agosto, M. Lansalot, B. Charleux, Controlled radical polymerization of styrene in miniemulsion mediated by PEO-based trithiocarbonate macromolecular RAFT agents, Polym. Chem. 2 (2011) 355–362.
[12] A. Butté, G. Storti, M. Morbidelli, Miniemulsion Living Free Radical Polymerization by RAFT, Macromolecules 34 (17) (2001) 5885–5896.
[13] Y. Luo, X. Liu, Reversible addition–fragmentation transfer (RAFT) copolymerization of methyl methacrylate and styrene in miniemulsion, J. Polym. Sci., Part A: Polym. Chem. 42 (24) (2004) 6248–6258.
[14] B.T.T. Pham, D. Nguyen, C.J. Ferguson, B.S. Hawkett, A.K. Serelis, C.H. Such, Miniemulsion Polymerization Stabilized by Amphipathic Macro RAFT Agents, Macromolecules 36 (24) (2003) 8907–8909.
[15] J.P. Russum, C.W. Jones, F.J. Schork, Continuous Living Polymerization in Miniemulsion Using Reversible Addition Fragmentation Chain Transfer (RAFT) in a Tubular Reactor, Ind. Eng. Chem. Res. 44 (8) (2005) 2484–2493.
[16] A.J.P. Van Zyl, R.F.P. Bosch, J.B. McLeary, R.D. Sanderson, B. Klumperman, Synthesis of styrene-based liquid-filled polymeric nanocapsules by the use of RAFT-mediated polymerization in miniemulsion, Polymer 46 (11) (2005) 3607–3615.
[17] C. Farcet, M. Lansalot, B. Charleux, R. Pirri, J.P. Vairon, Mechanistic Aspects of Nitroxide-Mediated Controlled Radical Polymerization of Styrene in Miniemulsion, Using a Water-Soluble Radical Initiator, Macromolecules 33 (23) (2000) 8559–8570.
[18] G. Pan, E.D. Sudol, V.L. Dimonie, M.S. El-Aasser, Nitroxide-Mediated Living Free Radical Miniemulsion Polymerization of Styrene, Macromolecules 34 (3) (2001) 481–488.
[19] M.F. Cunningham, M. Xie, K.B. McAuley, B. Keoshkerian, M.K. Georges, Nitroxide-Mediated Styrene Miniemulsion Polymerization, Macromolecules 35 (1) (2002) 59–66.
[20] C. Farcet, J. Belleney, B. Charleux, R. Pirri, Structural Characterization of Nitroxide-Terminated Poly(n-butyl acrylate) Prepared in Bulk and Miniemulsion Polymerizations, Macromolecules 35 (13) (2002) 4912–4918.
[21] R.A. Olson, M.E. Lott, J.B. Garrison, C.L.G. Davidson IV, L. Trachsel, D.I. Pedro, W. G. Sawyer, B.S. Sumerlin, Inverse Miniemulsion Photoiniferter Polymerization for the Synthesis of Ultrahigh Molecular Weight Polymers, Macromolecules 55 (19) (2022) 8451–8460.
[22] X. Qiao, L. Qiao, M. Zhou, X. Zhang, G. Shi, Y. He, E. Bourgeat-Lami, X. Pang, Carbon Quantum Dot-Catalyzed, Highly Efficient Miniemulsion Atom Transfer Radical Polymerization Induced by Visible Light, ACS Macro Lett. 11 (11) (2022) 1298–1305.
[23] Y. Wang, L. Fu, K. Matyjaszewski, Enzyme-Deoxygenated Low Parts per Million Atom Transfer Radical Polymerization in Miniemulsion and Ab Initio Emulsion, ACS Macro Lett. 7 (11) (2018) 1317–1321.
[24] S. Yu, T. Zhang, J. Xing, A facile approach preparing PMMA nanospheres through in-situ surfactant miniemulsion photopolymerization under green LED irradiation, Polymer 249 (2022), 124838.
[25] K. Jung, J. Xu, P.B. Zetterlund, C. Boyer, Visible-Light-Regulated Controlled/Living Radical Polymerization in Miniemulsion, ACS Macro Lett. 4 (10) (2015) 1139–1143.
[26] K. Jung, C. Boyer, P.B. Zetterlund, RAFT iniferter polymerization in miniemulsion using visible light, Polym. Chem. 8 (27) (2017) 3965–3970.
[27] Y. Wang, S. Dadashi-Silab, F. Lorandi, K. Matyjaszewski, Photoinduced atom transfer radical polymerization in ab initio emulsion, Polymer 165 (2019) 163–167.
[28] J. Mosnacek, A. Eckstein-Andicsova, K. Borska, Ligand effect and oxygen tolerance studies in photochemically induced copper mediated reversible deactivation radical polymerization of methyl methacrylate in dimethyl sulfoxide, Polym. Chem. 6 (13) (2015) 2523–2530.
[29] Q. Yang, J. Lalevee, J. Poly, Development of a Robust Photocatalyzed ATRP Mechanism Exhibiting Good Tolerance to Oxygen and Inhibitors, Macromolecules 49 (20) (2016) 7653–7666.
[30] X. Pan, S. Lathwal, S. Mack, J. Yan, S.R. Das, K. Matyjaszewski, Automated Synthesis of Well-Defined Polymers and Biohybrids by Atom Transfer Radical Polymerization Using a DNA Synthesizer, Angew. Chem., Int. Ed. 56 (10) (2017) 2740–2743.
[31] L.A. Navarro, A.E. Enciso, K. Matyjaszewski, S. Zauscher, Enzymatically Degassed Surface-Initiated Atom Transfer Radical Polymerization with Real-Time Monitoring, J. Am. Chem. Soc. 141 (7) (2019) 3100–3109.
[32] G. Gody R. Barbey M. Danial S. Perrier Ultrafast RAFT polymerization: multiblock copolymers within minutes Polym. Chem. 6 (9) (2015) 1502-1151.
[33] R. Chapman, A.J. Gormley, M.H. Stenzel, M.M. Stevens, Combinatorial Low-Volume Synthesis of Well-Defined Polymers by Enzyme Degassing, Angew. Chem., Int. Ed. 55 (14) (2016) 4500–4503.
[34] S. Shanmugam, J. Xu, C. Boyer, Aqueous RAFT Photopolymerization with Oxygen Tolerance, Macromolecules 49 (24) (2016) 9345–9357.
[35] I.H. Lee, E.H. Discekici, A. Anastasaki, J.R. de Alaniz, C.J. Hawker, Controlled radical polymerization of vinyl ketones using visible light, Polym. Chem. 8 (21) (2017) 3351–3356.
[36] Z. Liu, Y. Lv, Z. An, Enzymatic Cascade Catalysis for the Synthesis of Multiblock and Ultrahigh-Molecular-Weight Polymers with Oxygen Tolerance, Angew. Chem., Int. Ed. 56 (44) (2017) 13852–13856.
[37] G. Ng, J. Yeow, J. Xu, C. Boyer, Application of oxygen tolerant PET-RAFT to polymerization-induced self-assembly, Polym. Chem. 8 (18) (2017) 2841–2851.
[38] J. Yeow, R. Chapman, J. Xu, C. Boyer, Oxygen tolerant photopolymerization for ultralow volumes, Polym. Chem. 8 (34) (2017) 5012–5022.
[39] S.E. Seo, E.H. Discekici, Y. Zhang, C.M. Bates, C.J. Hawker, Surface-initiated PET-RAFT polymerization under metal-free and ambient conditions using enzyme degassing, J. Polym. Sci. 58 (1) (2020) 70–76.
[40] G. Ng, S.W. Prescott, A. Postma, G. Moad, C.J. Hawker, C. Boyer, Strategies for Achieving Oxygen Tolerance in Reversible Addition-Fragmentation Chain Transfer Polymerization, Macromol. Chem. Phys. 2300132 (2023).
[41] X. Dong, L. Wang, Y. He, Z. Cui, P. Fu, M. Liu, X. Qiao, G. Shi, X. Pang, Simple and robust nitroxide-mediated polymerization with oxygen tolerance, Polym. Chem. 12 (48) (2021) 7010–7015.

[42] N. Ballard, M. Aguirre, A. Simula, A. Agirre, J.R. Leiza, J.M. Asua, S. Van Es, New Class of Alkoxyamines for Efficient Controlled Homopolymerization of Methacrylates, ACS Macro Lett. 5 (9) (2016) 1019–1022.
[43] N. Ballard, A. Simula, M. Aguirre, J.R. Leiza, S. van Es, J.M. Asua, Synthesis of poly (methyl methacrylate) and block copolymers by semi-batch nitroxide mediated polymerization, Polym. Chem. 7 (45) (2016) 6964–6972.
[44] H. Ma, L. Wang, Y. Liang, Z. Cui, P. Fu, M. Liu, X. Qiao, X. Pang, Novel tetraphenylethylene (TPE)-functionalized nitroxide/alkoxyamine for nitroxide-mediated homogeneous and heterogeneous polymerizations, Polym. Chem. 12 (4) (2021) 526–533.
[45] M. Cunninghama, M. Lina, C. Buraginaa, S. Miltona, D. Nga, C.C. Hsua, B. Keoshkerian, Maximizing polymer livingness in nitroxide-mediated miniemulsion polymerizations, Polymer 46 (2005) 1025–1032.